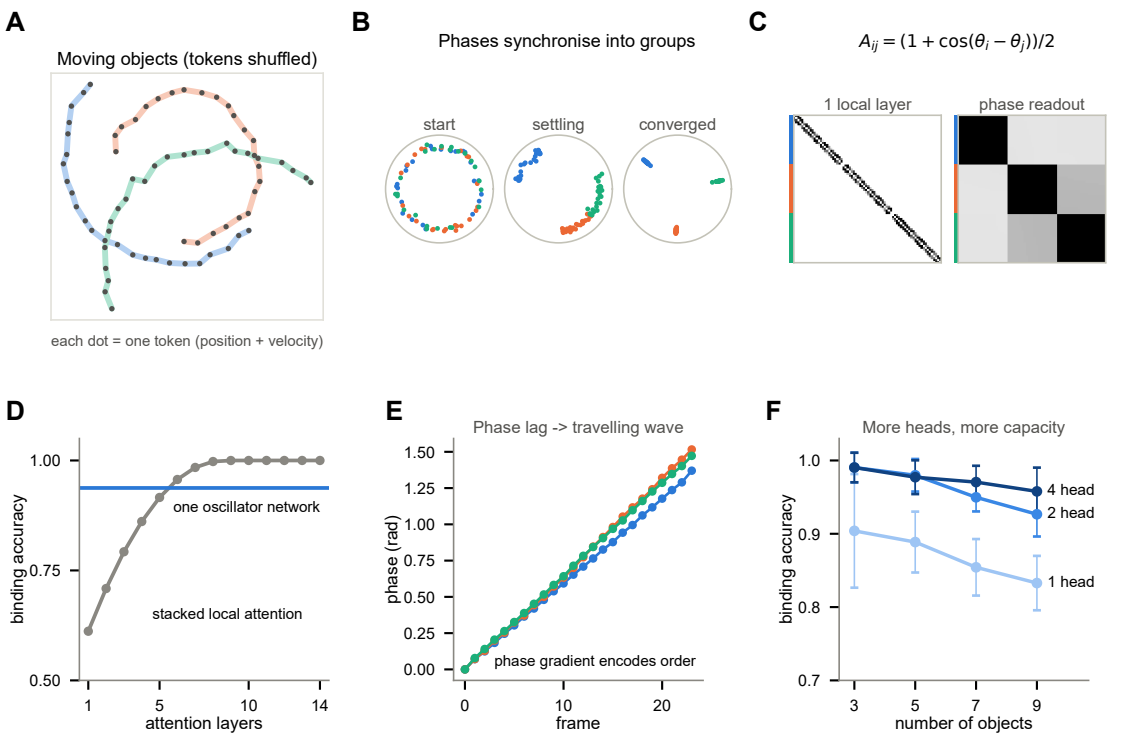

A
Moving objects (tokens shuffled)
each dot = one token (position + velocity)
B
Phases synchronise into groups
start
settling
converged
C
$A_{ij} = (1 + \cos(\theta_i - \theta_j))/2$
1 local layer
phase readout
D
binding accuracy
1.00
0.75
0.50
one oscillator network
stacked local attention
1
5
10
14
attention layers
E
Phase lag -> travelling wave
phase (rad)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
phase gradient encodes order
0
10
20
frame
F
More heads, more capacity
binding accuracy
1.0
0.9
0.8
0.7
4 head
2 head
1 head
3
5
7
9
number of objects